

Sampad Kumar Kar

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EDUCATION

- **Chennai Mathematical Institute** Chennai, India
M.Sc. - Computer Science; GPA: 8.94; transcript
Notable Coursework: Advanced Machine Learning, Reinforcement Learning, Mathematical Statistics, Computer Vision, Natural Language Processing
2022 - 2024
- **Chennai Mathematical Institute** Chennai, India
B.Sc.(Hons.) - Mathematics and Computer Science; GPA: 8.82; transcript
Notable Coursework: Calculus, Differential Equations, Probability Theory, Linear Algebra, Theoretical Foundations of ML, Data Mining and ML, Probability and Statistics with R, Linear Programming and Combinatorial Optimisation
2019 - 2022

EXPERIENCE

- **Michelin** Pune (Hybrid), India
Data Science Intern
Dec 2023 - May 2024
 - **Projects:** TireScan
 - **Contribution:** Developed TireScan, a system for extracting tire information from web data using LLMs. Implemented and compared four algorithms, enhancing accuracy and efficiency. Documented project methodology and results, highlighting the RAG pipeline for its high accuracy. *Report*
- **BNY Mellon** Chennai, India
Financial Modelling Intern
May 2023 - July 2023
 - **Project:** Enhancing Short Term Cash Flow Model
 - **Contribution:** Improved BNY Mellon's STCF forecasting by introducing quantitative methods for identifying seasonal patterns, enhancing accuracy across LOBs and currencies. Conducted statistical tests and modeling to validate the quantitative framework, which was subsequently submitted to Risk for audit and adoption. *Report*
- **Trumpf** Chennai, India
Computer Vision Intern
Nov 2022 - Apr 2023
 - **Project:** Sheet Metal Part Identification
 - **Contribution:** Developed a deployable pipeline for sheet metal part identification, using transfer learning on pre-trained models to classify images into 10 predefined categories and output the corresponding CAD files. *Report*
- **Launchpad.ai** Remote
Machine Learning Intern
July 2022 - Sep 2022
 - **Project:** Fitness Activity Recognition (in collaboration with Nike)
 - **Contribution:** Developed a pipeline for comparing trainee and expert fitness videos. Utilized 'Dynamic Time Warping' on coordinates extracted via 'Mediapipe BlazePose' to handle variations in camera angles. *Report*
- **Cheenta** Remote
Mathematics Instructor (part-time)
Oct 2019 - Apr 2022
 - **Mathematics Olympiad Instructor:** Taught various courses for Indian National Mathematics Olympiads on various topics like Geometry, Number Theory, Algebra etc.

NOTABLE PROJECTS

- **Clustering - Fast Text based Clustering:** Developed four clustering techniques, including two optimized versions of sklearn's KMeans and two novel techniques designed from scratch to efficiently cluster large datasets using 'Jaccard Similarity' as the metric. *GitHub* (April '22)
- **Computer Vision - Facial Emotion Detector:** Created a real-time facial expression detector using Deep Residual Networks upon the *Kaggle Facial Expression Dataset*, with deployment for live webcam-based emotion recognition. *GitHub* (Jan '23)
- **Large Language Models - Local GPT for Documents:** Developed a local, internet-independent framework for querying and chatting with users' *Private Documents* in natural language using LLMs, while also identifying document references from the knowledge base. *GitHub* (Aug '22)

NOTABLE REPORTS

- **Master's Thesis (LLM) - SSMs: An Efficient Alternative to Attention:** This thesis investigates state space models (SSMs) as alternatives to Transformers for deep sequence modeling. It examines the efficiency and performance of SSM variants, including S4, Mamba, LRUs, and Griffin, through experiments on text, image, and audio generation tasks using tiny language models. *GitHub Report* (May '24)
- **ANN - Comparison of Regularization Techniques in DNNs:** (Reading Project): Comparative research on several popular regularization techniques using real world weather dataset. Also tested this on their own test set, to validate the conclusions based on training and validation errors, to come up with the best regularization paradigm. *Report* (Jan '22)

TA EXPERIENCE

- **Chennai Mathematical Institute** Chennai, India
Teaching Assistant *Jan 2023 - Apr 2024*
 - **Data Mining and Machine Learning:** Instructors: Prof. Madhavan Mukund, Prof. Pranabendu Misra
 - **Advanced Machine Learning:** Instructors: Prof. Pranabendu Misra
 - **Natural Language Processing:** Instructors: Prof. Ramaseshan Ramachandran
- **Sai University** Chennai, India
Teaching Assistant *Aug 2023 - Dec 2023*
 - **Introduction to Cryptography:** Instructors: Prof. Kavita Sutar

HONORS AND AWARDS

- Shriram Scholarship (Full Tution Fee Waiver and Stipend at CMI for Masters) - 2022 - present
- Shriram Scholarship (Full Tution Fee Waiver and Stipend at CMI for Bachelors) - 2019 - 2022
- NTSE Scholar - 2017
- INMO Merit Awardee - 2017
- KVPY Qualifier - 2018
- NSEP, NSEA, RMO Qualifier - 2017-18
- JEE Advanced AIR **4062**, JEE Mains AIR **3042** - 2019

SKILLS

- **Languages:** Python, R, Matlab C++, Java, Haskell
- **ML Frameworks:** NumPy, Pandas, Scikit, Tensorflow, NLTK, OpenCV, Matplotlib, Seaborn
- **Cloud Frameworks:** AWS, Azure, VertexAI
- **Web Dev:** HTML, CSS, Bootstrap, Javascript, Flask, Jinja2, SQLAlchemy
- **Tools:** Git, PostgreSQL, L^AT_EX, Excel, SQLite
- **Platforms:** Linux, Windows, Mac
- **Soft Skills:** Leadership, Event Management, Writing, Public Speaking, Time Management